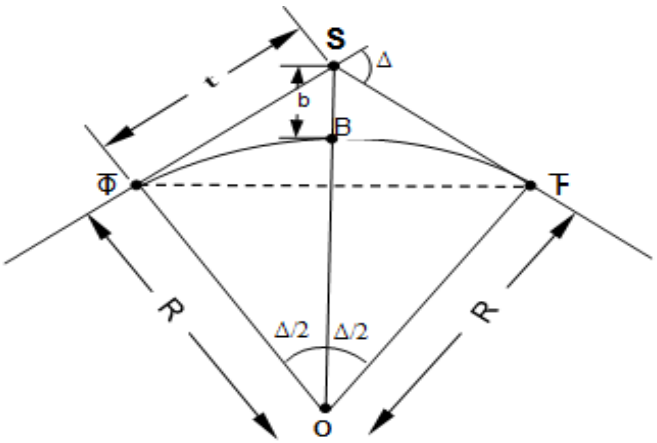
Highway Engineering Problem Session-3

Question - 1: Find all components of a horizontal curve with a radius of 300 m and curve angle of 90º.



Solution:

R=300m; $\Delta = 90^0$; t=? ; D= ? ; b= ?

$t = \mathrm{R} * tg(\frac{\Delta}{2})$ $t = 300 * tg(\frac{90}{2}) = 300$ m

$$\mathrm{D} = 2 * \pi * \mathrm{R} * \frac{\Delta}{360} = 2 * \pi * 300 * \frac{(90)}{360} = 471.24 \text{ m}$$

$$b = R * \left(\frac{1}{\cos\left(\frac{\Delta}{2}\right)} - 1 \right) = 300 * \left(\frac{1}{\cos\left(\frac{90}{2}\right)} - 1 \right) = 124.26m$$

Question - 2: Data of a horizontal curve is given below. Find missing components of the horizontal curve and find km values for the other critical points of the horizontal curve.

Given: Supplementary angle: 150° ; tangent length: 280.007 m; $\Phi_{km} = 1+550$

Solution:

Curve angle Δ = 180-150 = 30°

$$t = R * tg\left(\frac{\Delta}{2}\right)$$

$$280.007 = R * tg\left(\frac{30}{2}\right) \Rightarrow \text{R = 1045 m.}$$

$$D = \frac{2\pi R}{360} * \Delta$$

$$\mathrm{D} = \frac{2 * \pi * 1045}{360} * (30) = 547.16 \text{ m.}$$

$$b = R * (\frac{1}{\cos\frac{\Delta}{2}} - 1)$$

$$\mathrm{b} = 1045 * (\frac{1}{\cos(\frac{30}{2})} - 1) = 36.86 \text{ m.}$$

Other critical points' km values;

$T_{km} = \Phi_{km} + D = [(1+550) + 547.16] = 2+097^{16}$

Bisector point divides the arc length into two equal sections.

$$B_{km} = \Phi_{km} + \frac{D}{2} = [(1+550) + \frac{547.16}{2}] = 1+823^{58}$$

Question - 3: Data about two inverse horizontal curves designed on a two-way double lane highway.

a) Find all components of the horizontal curves.

b) Calculate the distance between the inverse horizontal curves and check if it is enough.

Data: $\Delta_1 = (59.73)^o$, $\Delta_2 = (68.76)^o$, Φ_{1km}=1+000, T_{1km}=1+344, Φ_{2km}=1+500, R_2=250 m.

Solution:

a) Components for Curve 1;

$D_1 = T_{1km} - \Phi_{1km} = (1 + 344) – (1 + 000) = 344$m.

$$D_1 = 2\pi R_1 * \frac{\Delta_1}{360} \quad ; \; 344 = 2\pi R_1 * \frac{(59.73)}{360} \quad ; \quad R_1 = 330 \text{ m}$$

$$t_1 = R_1 * tg(\frac{\Delta_1}{2}) = 330 * tg(\frac{59.73}{2}) = 189.49 \text{ m}$$

$$b_1 = R_1 * (\frac{1}{\cos(\frac{\Delta_1}{2})} - 1) = R_1 * (\sec(\frac{\Delta_1}{2}) - 1) = 330 * (\sec(\frac{59.73}{2}) - 1) = 50.53 \text{ m}$$

Components for Curve 2;

$$D_2 = 2\pi R_2 * \frac{\Delta_2}{360} = 2\pi 250 * \frac{(68.76)}{360} = 299.87 \text{ m}$$

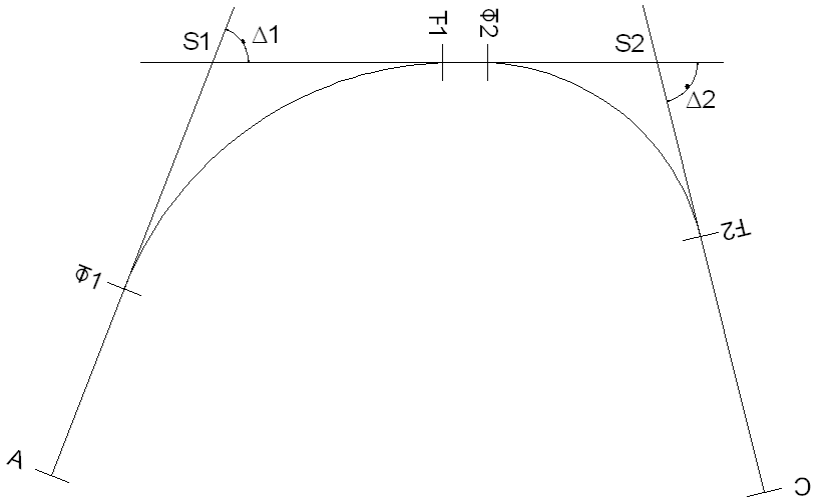
$$t_2 = R_2 * tg(\frac{\Delta_2}{2}) = 250 * tg(\frac{68.76}{2}) = 171.05 \text{ m}$$

$$b_2 = R_2 * (\frac{1}{\cos(\frac{\Delta_2}{2})} - 1) = R_2 * (\sec(\frac{\Delta_2}{2}) - 1) = 250 * (\sec(\frac{68.76}{2}) - 1) = 52.92 \text{ m}$$

b) $l = \Phi_{2km} - T_{1km}$ =(1+500)-(1+344)=156 m > 60 m therefore, it is enough.

Question - 4: As presented in the figure below, a route section is designed with two curves. The radius of the first curve is R_1 = 400m Δ_1= (69)°; for the second curve, the curve angle is Δ_2 = (75.26)°. According to the existing route geometry, it is planned to use the maximum radius possible for the second curve. The distance limit between the corresponding curves is 30 meters and the distances between the following points are given as $\overline{AS_1} = \overline{S_1S_2} = \overline{S_2C}$ =500 meters.

a) Find the missing components of the first curve
b) Find the radius of the second curve
c) Find the total length of the route section.



Solution:

a) $t_1 = \mathrm{R}_1 * tg\left(\frac{\Delta_1}{2}\right) = 400 * tg\left(\frac{69}{2}\right) = 274.91\,\mathrm{m}$

$$\mathrm{D}_1 = \frac{2\pi \mathrm{R}_1.\Delta_1}{360} = \frac{2\pi 400 * (69)}{360} = 481.71\,\mathrm{m}$$

$$\mathrm{b}_1 = \mathrm{R}_1 * \left(\frac{1}{\cos(\frac{\Delta}{2})} - 1\right) = 400 * \left(\frac{1}{\cos(\frac{69}{2})} - 1\right) = 85.36\,\mathrm{m}$$

b) $t_2 = \overline{S_1S_2} - t_1 - 30$

$t_2 = 500 - 274.91 - 30 = 195.09\,\mathrm{m}$

$t_2 = \mathrm{R}_2 * tg\left(\frac{\Delta_2}{2}\right) \Rightarrow 195.09 = \mathrm{R}_2 * tg\left(\frac{75.26}{2}\right) \Rightarrow \mathrm{R}_2 = 253.06 \text{ m} \Rightarrow \mathrm{R}_{2,yuv} = 250 \text{ m}$ taken. (the radius has to be a multiple of 5!)

$$t_{2,yeni} = \mathrm{R}_{2,yuv} * tg\left(\frac{\Delta_2}{2}\right) = 250 * tg\left(\frac{75.26}{2}\right) = 192.73 \text{ m}$$

$$\mathrm{D}_2 = \frac{2\pi 250 * 75.26}{360} = 328.38 \text{ m}$$

c)

A_{km} = 0+000

$\Phi_{1km} = A_{km} + \left(\overline{AS_1} - t_1\right)$ = A_{km} + (500-274.91) = $0+225^{09}$

$\Gamma_{1km} = \Phi_{1km} + D_1 = (0+225^{09}) + (481.71) = 0+706^{80}$

$\Phi_{2km} = \Gamma_{1km} + \left(\overline{S_1S_2} - t_1 - t_{2,yeni}\right) = (0+706^{80}) + (500-274.91-192.73) = 0+739^{16}$

$\Gamma_{2km} = \Phi_{2km} + D_2 = (0+739^{16}) + (328.38) = 1+067^{54}$

$C_{km} = \Gamma_{2km} + \left(\overline{CS_2} - t_{2,yeni}\right) = (1+067^{54}) + (500 - 192.73) = 1+374^{81}$

total length of the route = 1374.81 m.

Question - 5: A two-way double lane highway has a project velocity of 70 km/h. Some data about this highway and criteria is given below. There is a curve in this road section without a transition line and without a ramp. It is required to perform constant centerline superelevation on this curve.

Project Data	**Project Criteria**
$\Delta = (59.73)^o$	Platform width =10 m
Φ_{km} =1+000	Crown slope = 2% , Max Superelevation = 8%
Γ_{km} =1+344	Theorotical superelevation = 0.786*(V^2/R) (%), L_d = 0.0354*(V^3/R)

a) Find all components of the curve.
b) By considering that at most half of the centrifugal force can be compensated by the friction, is the curve radius proper for safely passsing?
c) Draw the constant centerline superelevation diagram for this curve and find the superelevation heights of the inner and the outer edges of the given cross sections. 50(0+990), Φ(1+000), 51(1+010).

Solution:

a) $D = \Gamma_{(km)} - \Phi_{(km)} = 1344 - 100 = 344m$

$$D = \frac{2\pi R\Delta}{360} \qquad 344 = \frac{2\pi R(59.73)}{360} \Rightarrow R = 330\,m$$

$$t = R\tan\left(\frac{\Delta}{2}\right) = 330 * \tan\left(\frac{59.73}{2}\right) = 189.49m$$

$$b = R\left(\frac{1}{\cos\left(\frac{\Delta}{2}\right)} - 1\right) = 330 * \left(\frac{1}{\cos\left(\frac{59.73}{2}\right)} - 1\right) = 50.53m$$

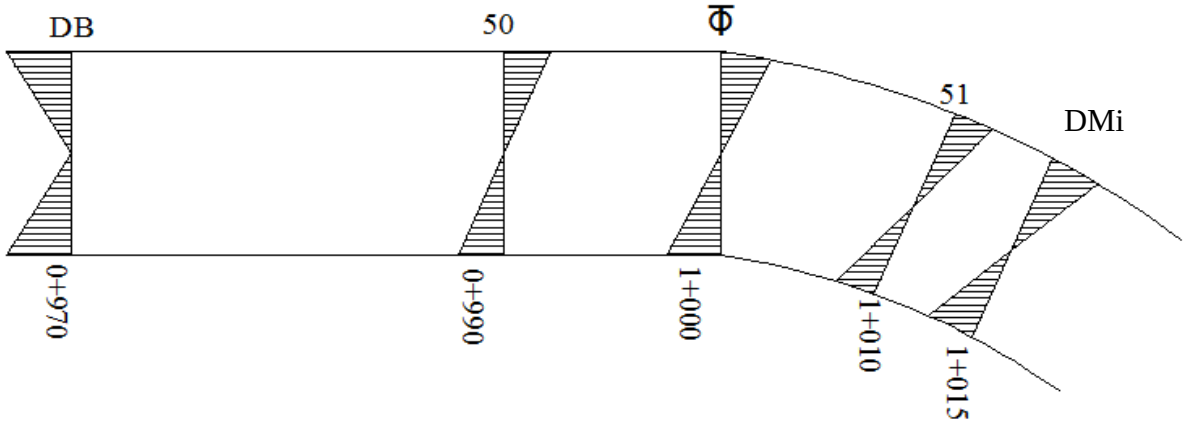
b) $q_{pra} = 0.5 * 0.786 * \frac{V_p^2}{R} \rightarrow 8 = 0.5 * 0.786 * \frac{70^2}{R_{\min}} \Rightarrow R_{\min} = 240\,m$ → Radius is proper.

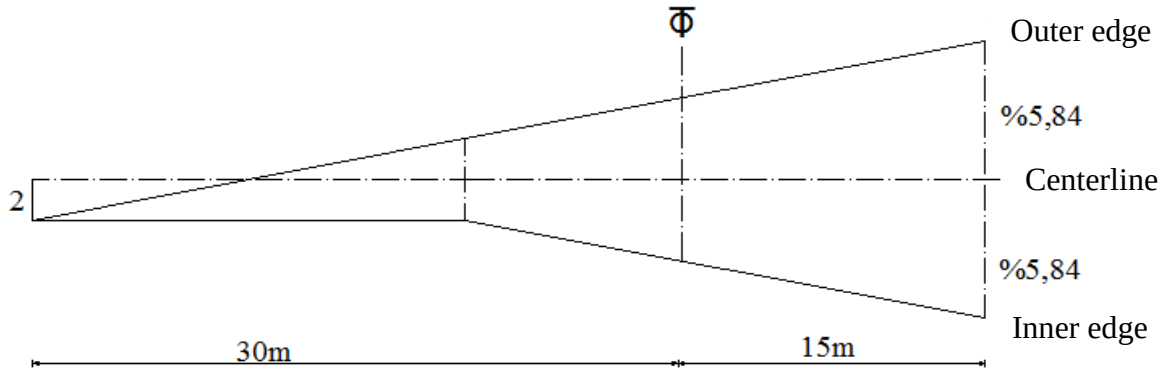
c) $q_{pra} = 0.5 * 0.786 * \frac{V_p^2}{R} = 0.786 * 0.5 * \frac{70^2}{330} = 5.84\% < 8\%$

$$L_d = 0.0354 * \frac{V_p^3}{R} = 0.0354 * \frac{70^3}{330} = 36.79m < 45m$$

$$L_d = 45m$$

No	Km	d(%)		h(m)	
		ie	oe	ie	oe
DB	0+970	2	2	0.10	0.10
50	0+990	2	1.48	0.10	0.074
Φ	1+000	3.23	3.23	0.162	0.162
51	1+010	4.97	4.97	0.249	0.249
DS	1+015	5.84	5.84	0.292	0.292





$$h_{\min} = \frac{b}{2} * Q_{\min} = \frac{10}{2} * 0.02 = 0.10m$$

$$h_{maks} = \frac{b}{2} * Q_{maks} = \frac{10}{2} * 0.0584 = 0.292\,m$$

$$\frac{5.84-(-2)}{45} = \frac{x-(-2)}{30} \Rightarrow x = 3.23\%$$

h_{Φ} = ((b/2)*Q_{Φ}) = (10/2)*0.0323 = 0.162m

$$\frac{5.84-(-2)}{45} = \frac{x-(-2)}{20} \Rightarrow x = 1.48\%$$

$$h_{50} = \frac{10}{2} * 0.0148 = 0.074\ \text{m}$$

$$\frac{5.84-(-2)}{45} = \frac{x-(-2)}{40} \Rightarrow x = 4.97\%$$

$$h_{51} = \frac{10}{2} * 0.0497 = 0.2485\ \text{m}$$